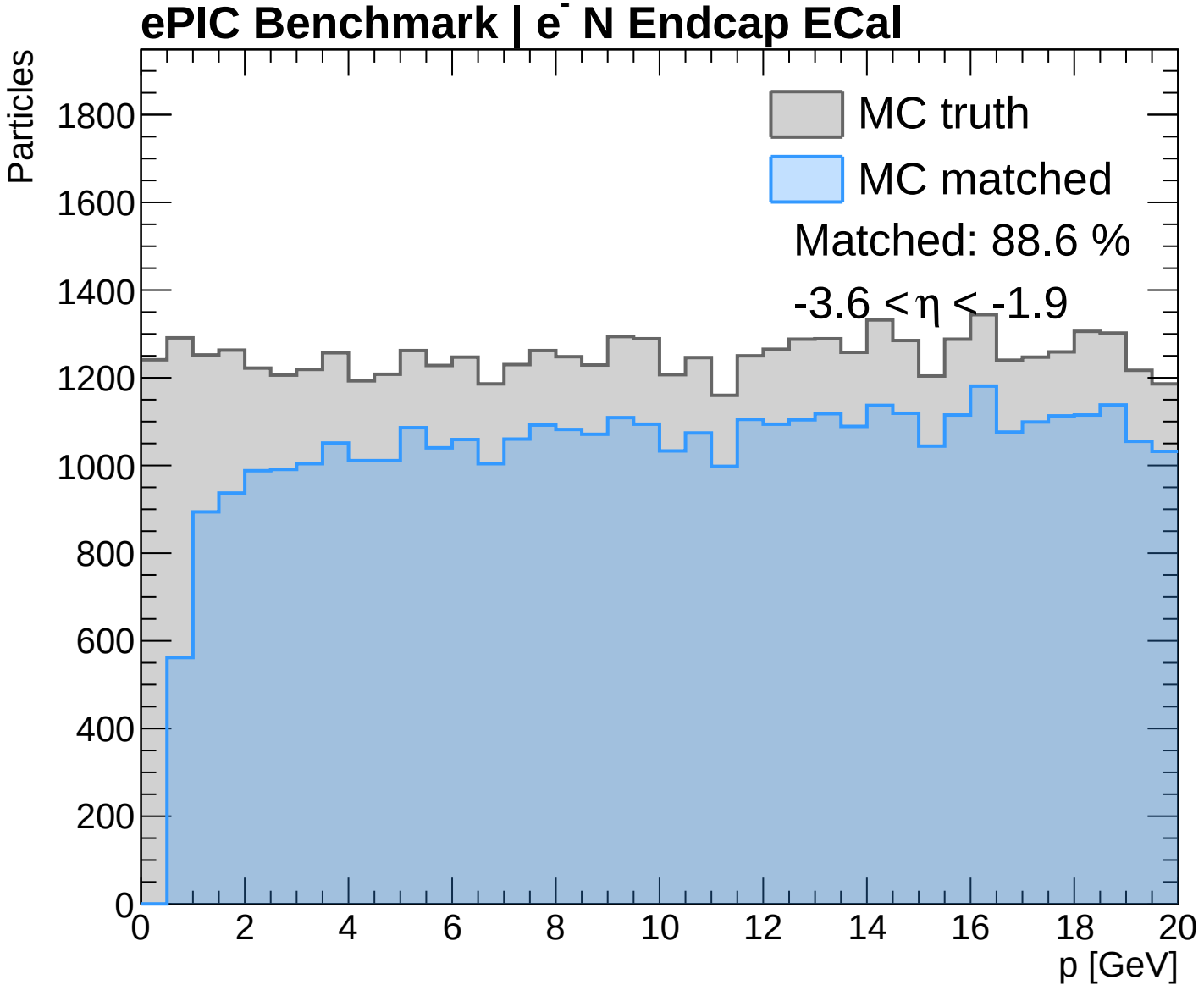
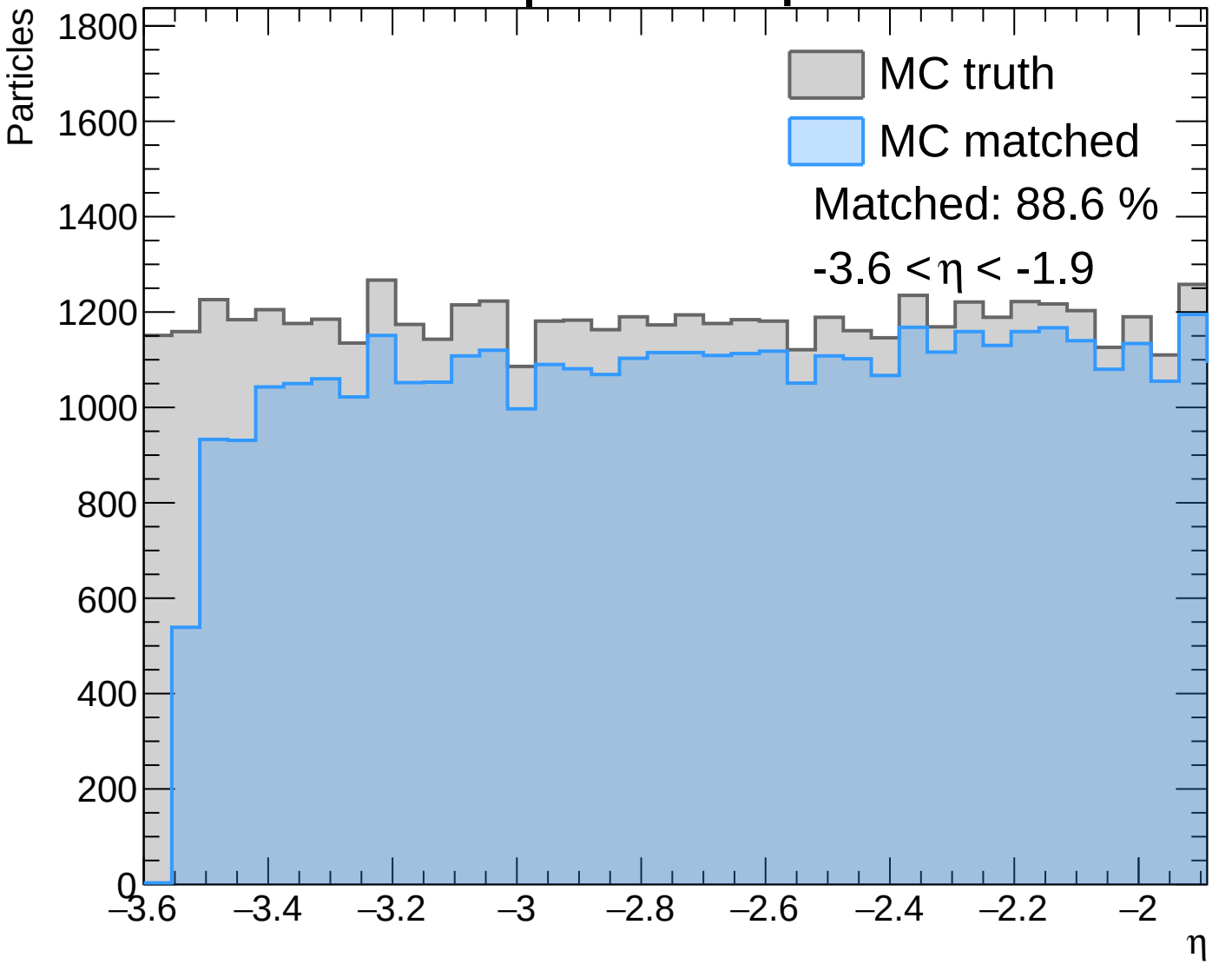
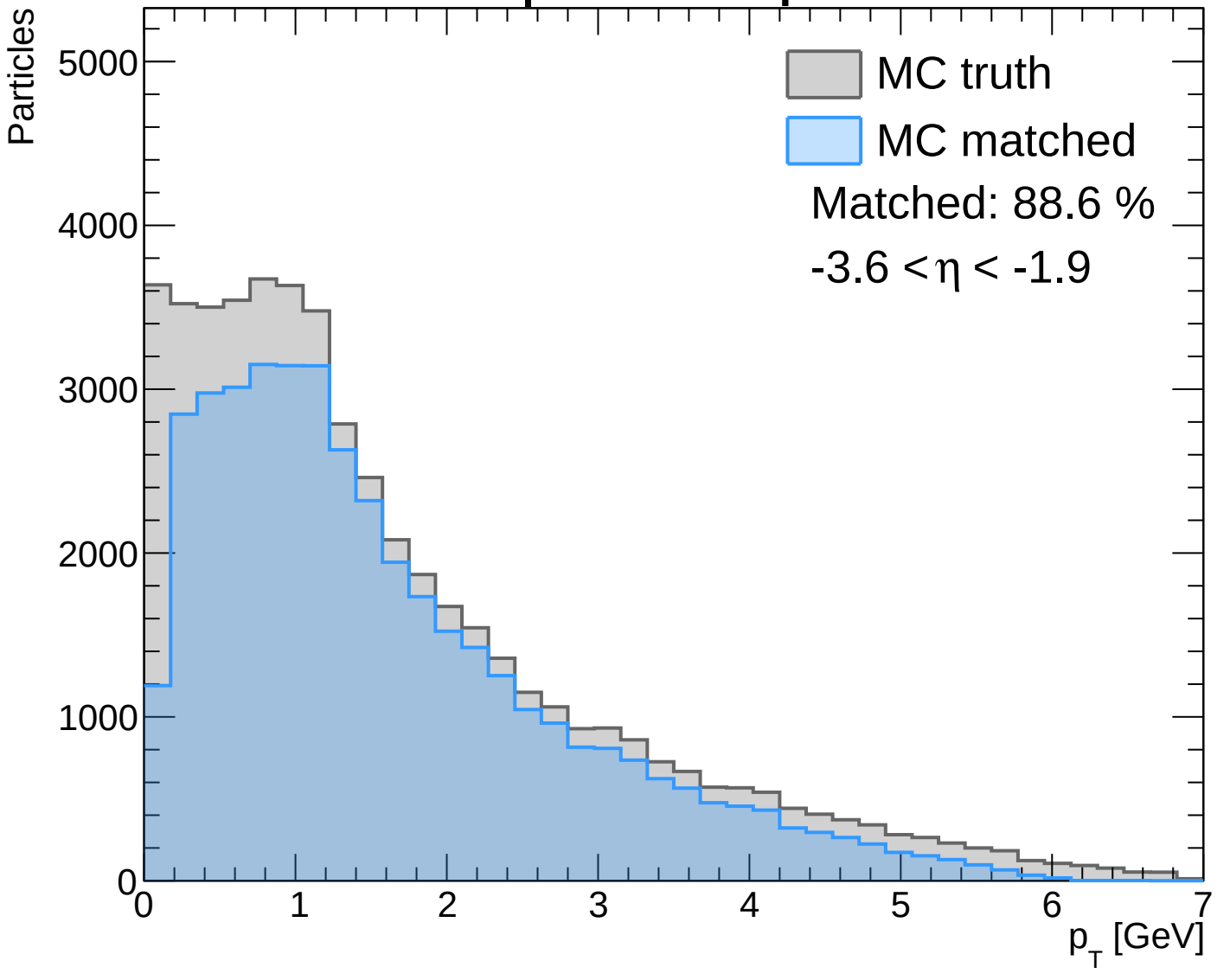
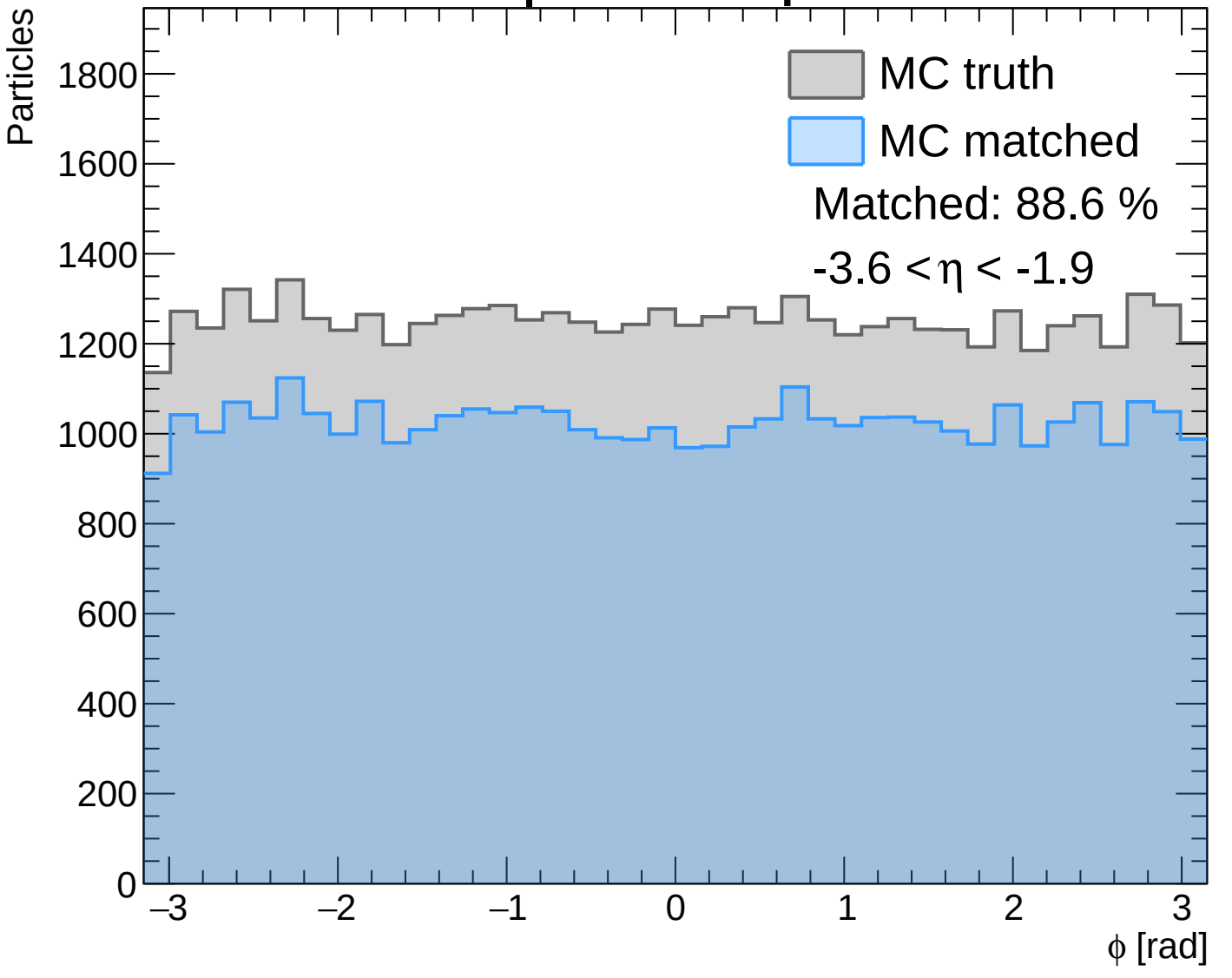
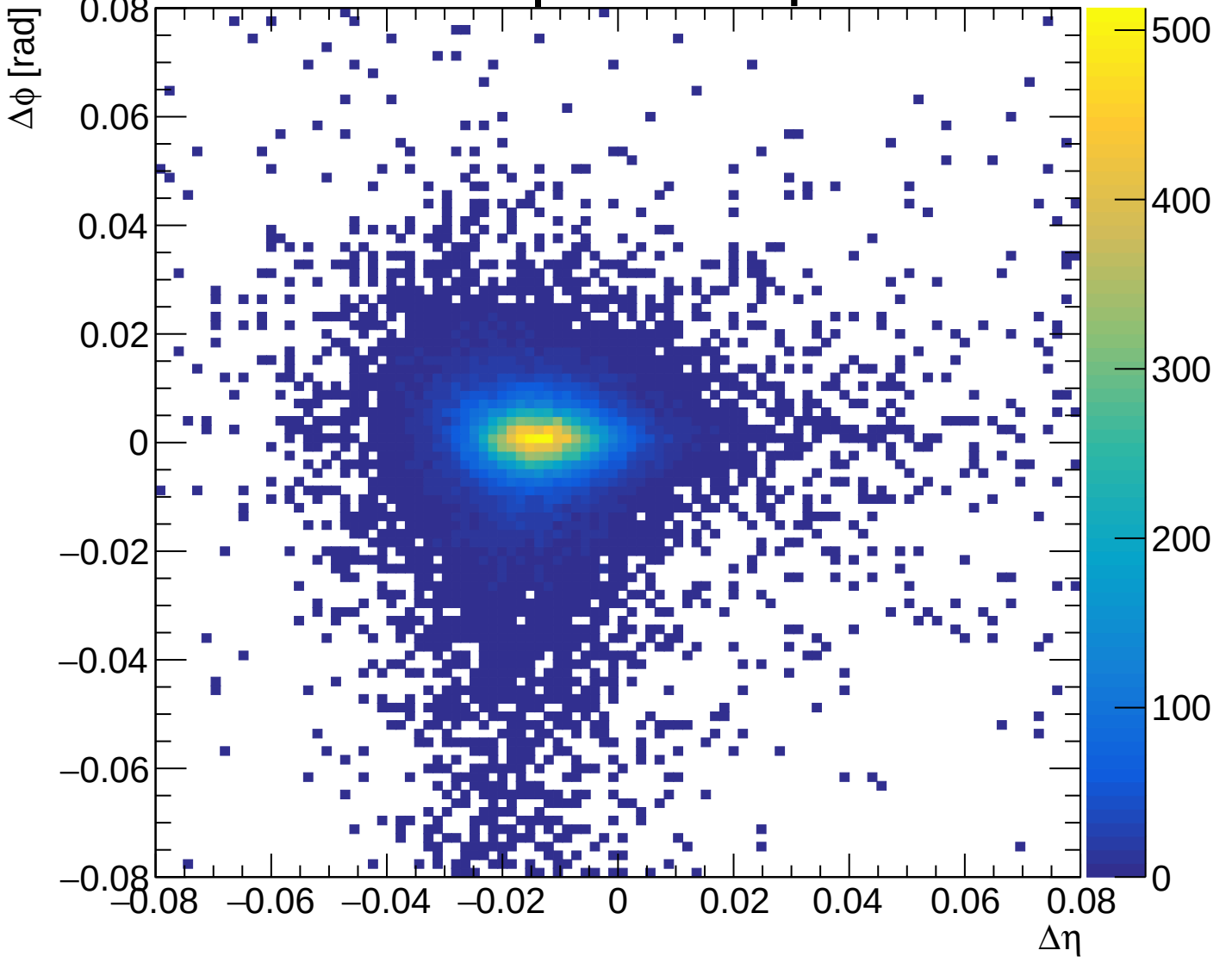
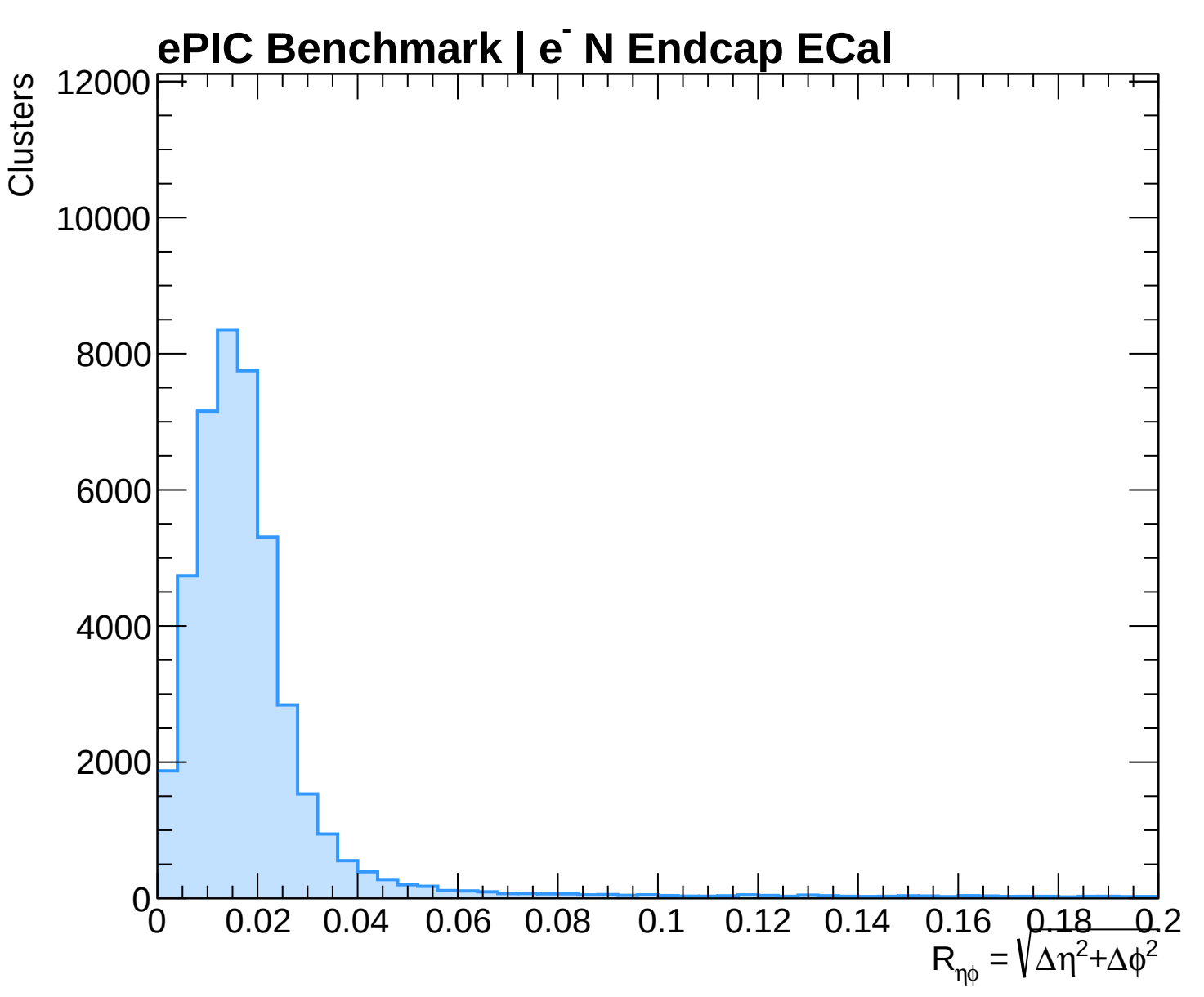

ePIC Benchmark | e- N Endcap ECal
Particles
MC truth
MC matched
Matched: 88.6 %
-3.6 < η < -1.9
0
200
400
600
800
1000
1200
1400
1600
1800
0
2
4
6
8
10
12
14
16
18
20
p [GeV]

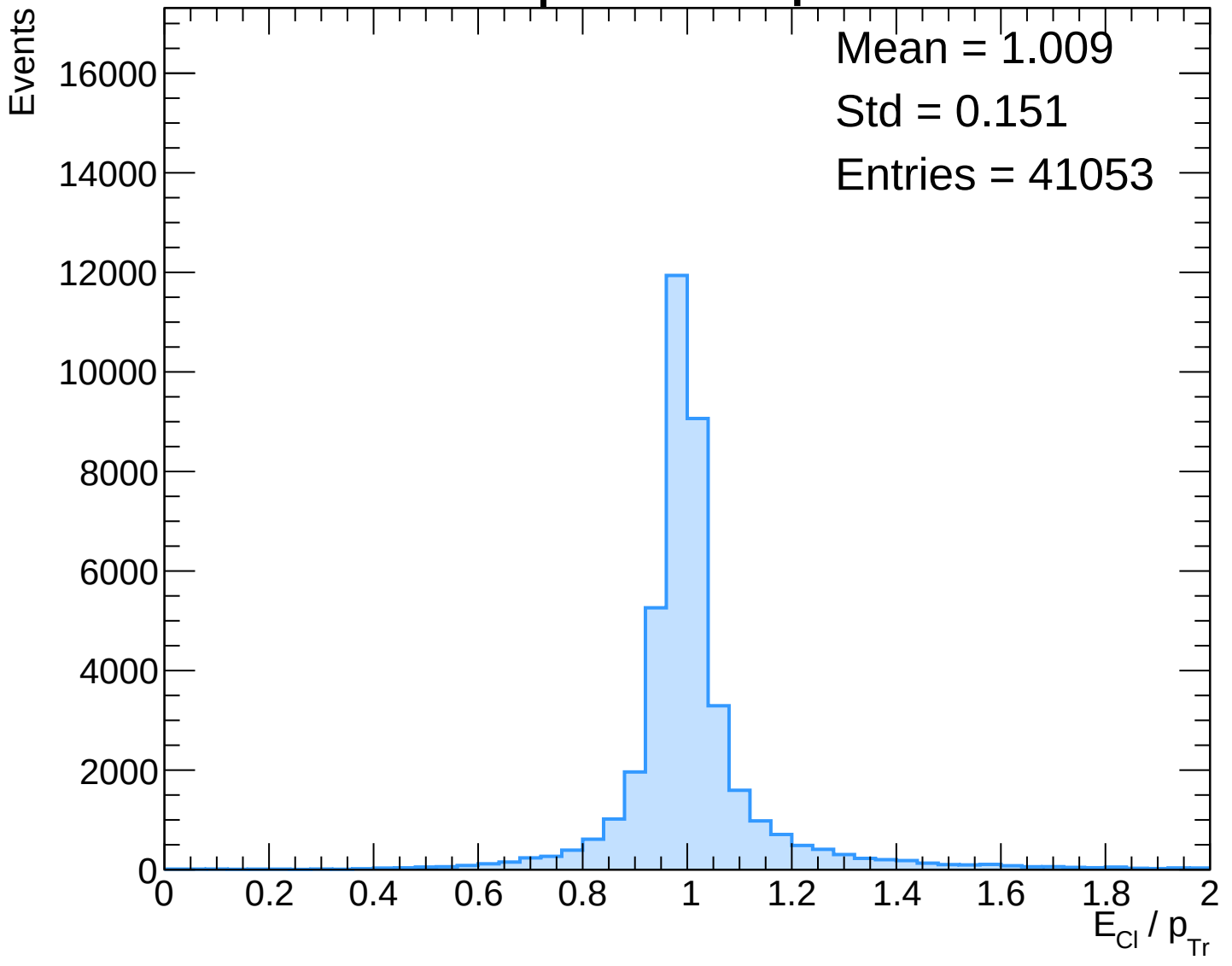

ePIC Benchmark | e⁻ N Endcap ECal
Particles
1800
1600
1400
1200
1000
800
600
400
200
0
MC truth
MC matched
Matched: 88.6 %
-3.6 < η < -1.9
–3.6
–3.4
–3.2
–3
–2.8
–2.6
–2.4
–2.2
–2
η

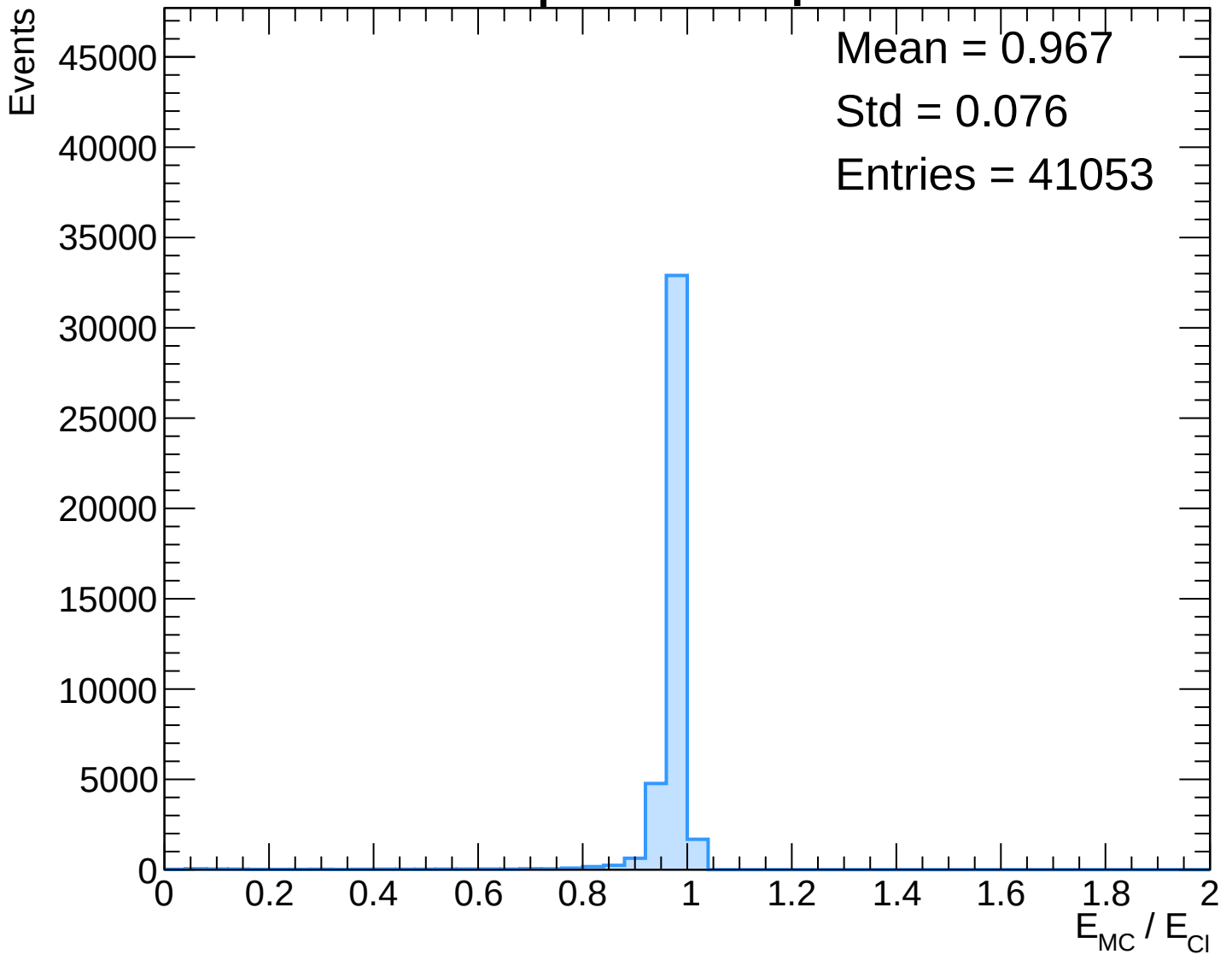

ePIC Benchmark | e- N Endcap ECal
Particles
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
6
7
pT [GeV]
MC truth
MC matched
Matched: 88.6 %
-3.6 < η < -1.9

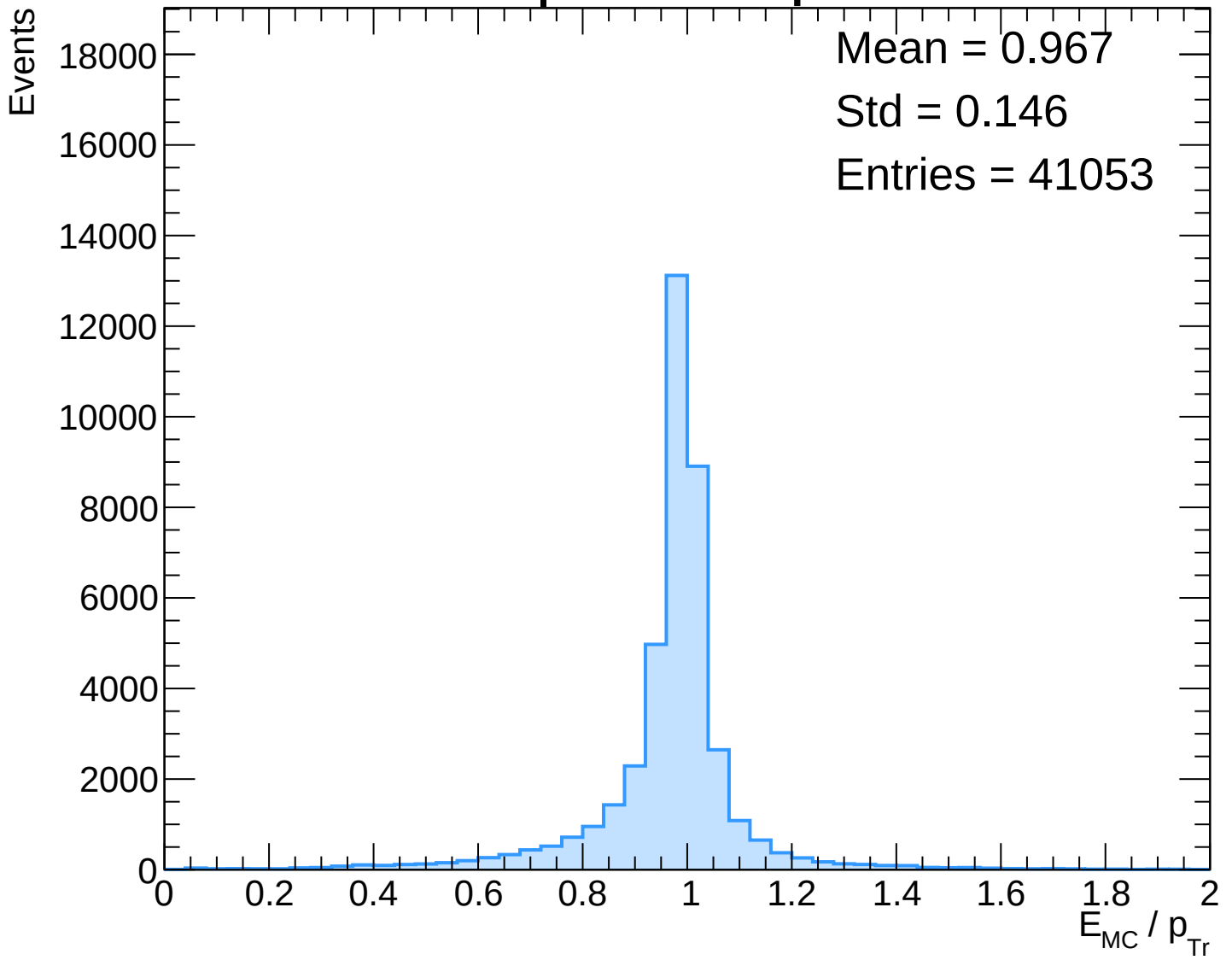

ePIC Benchmark | e⁻ N Endcap ECal
Particles
1800
1600
1400
1200
1000
800
600
400
200
0
–3
–2
–1
0
1
2
3
ϕ [rad]
MC truth
MC matched
Matched: 88.6 %
-3.6 < η < -1.9

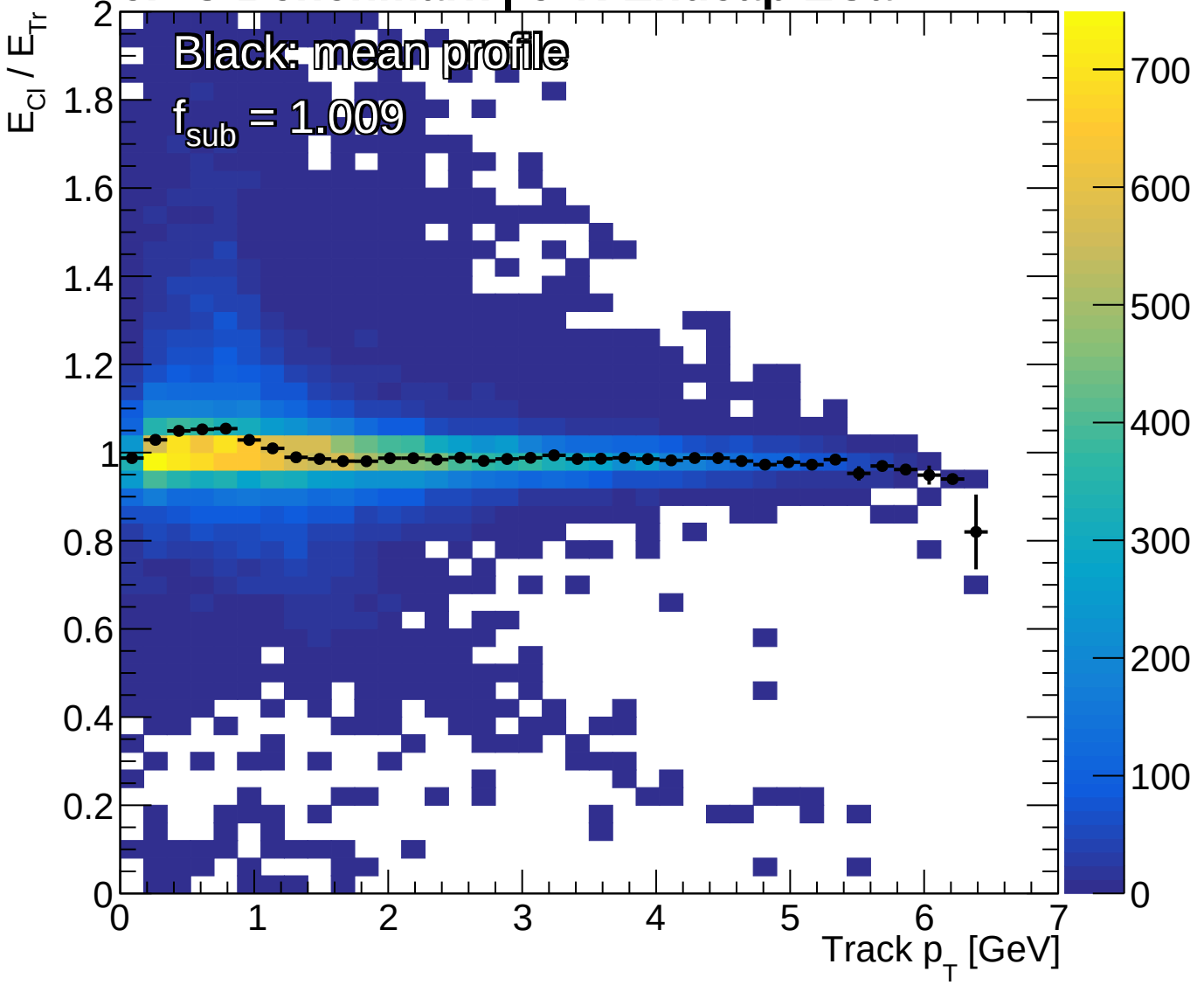

ePIC Benchmark | e⁻ N Endcap ECal
Δϕ [rad]
0.08
0.06
0.04
0.02
0
−0.02
−0.04
−0.06
−0.08
−0.08
−0.06
−0.04
−0.02
0
0.02
0.04
0.06
0.08
Δη
500
400
300
200
100
0

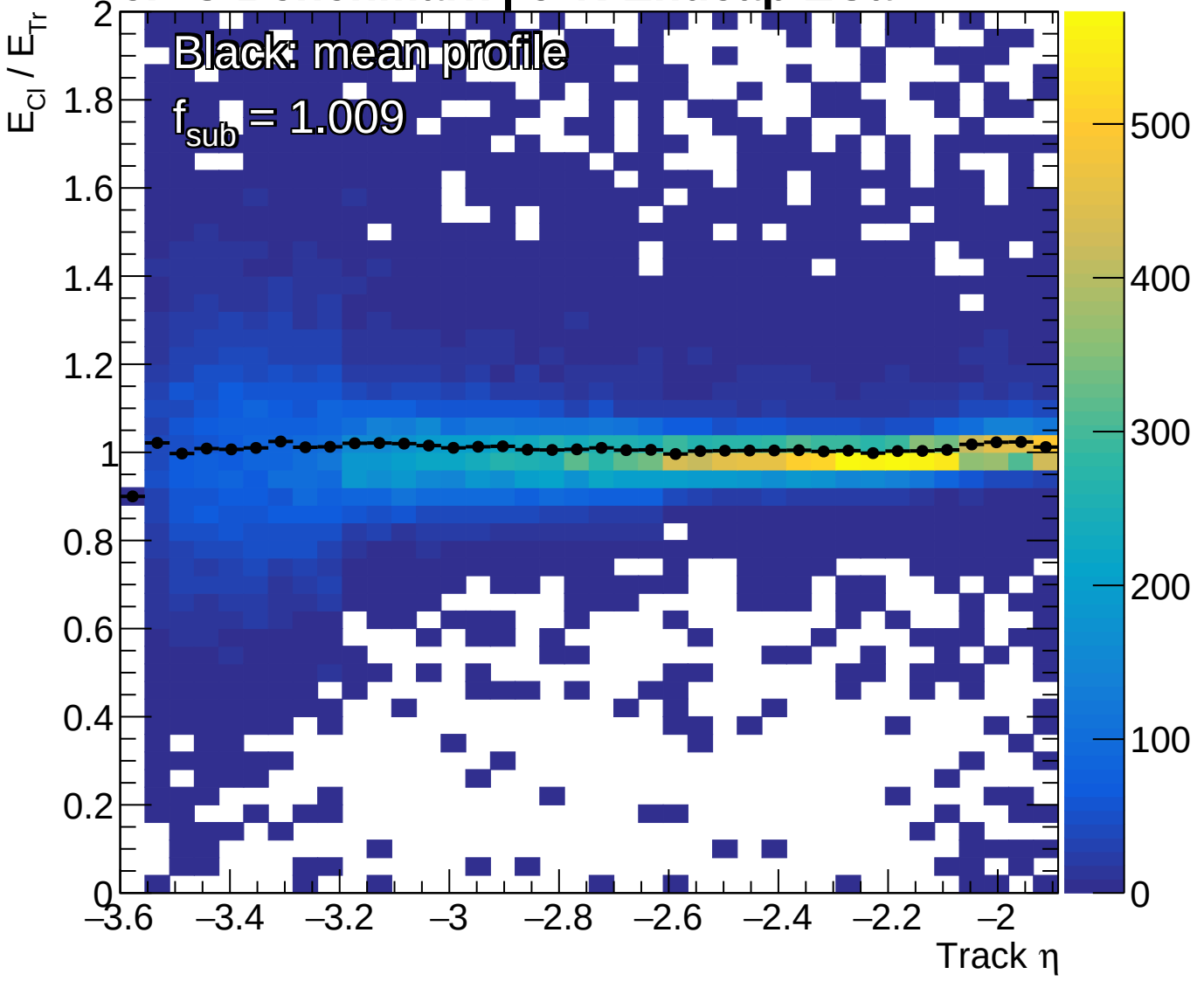

ePIC Benchmark | e- N Endcap ECal
Clusters
12000
10000
8000
6000
4000
2000
0
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
$R_{\eta\phi} = \sqrt{\Delta\eta^2+\Delta\phi^2}$

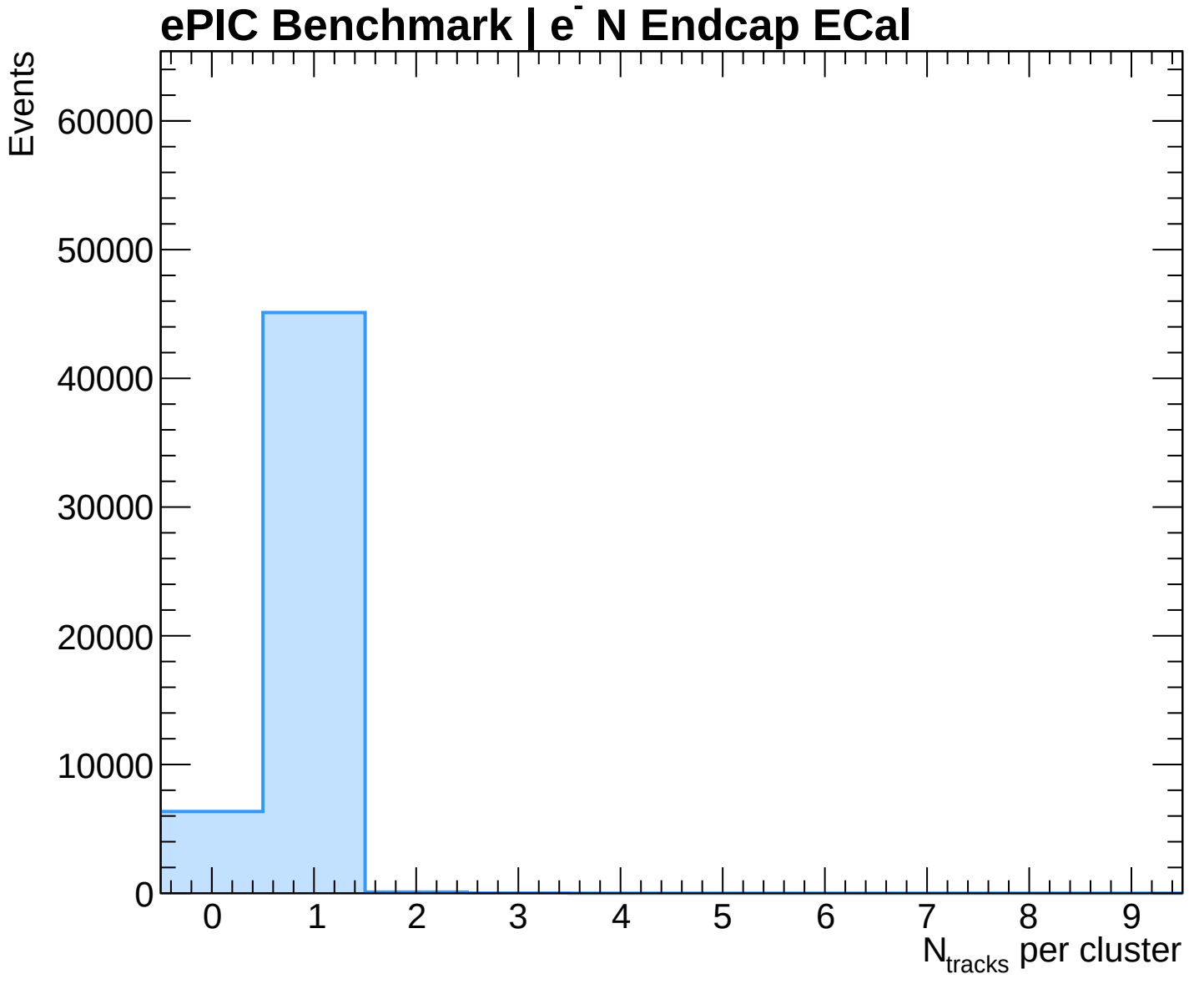

ePIC Benchmark | e- N Endcap ECal
Mean = 1.009
Std = 0.151
Entries = 41053
Events
16000
14000
12000
10000
8000
6000
4000
2000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_Cl / p_Tr

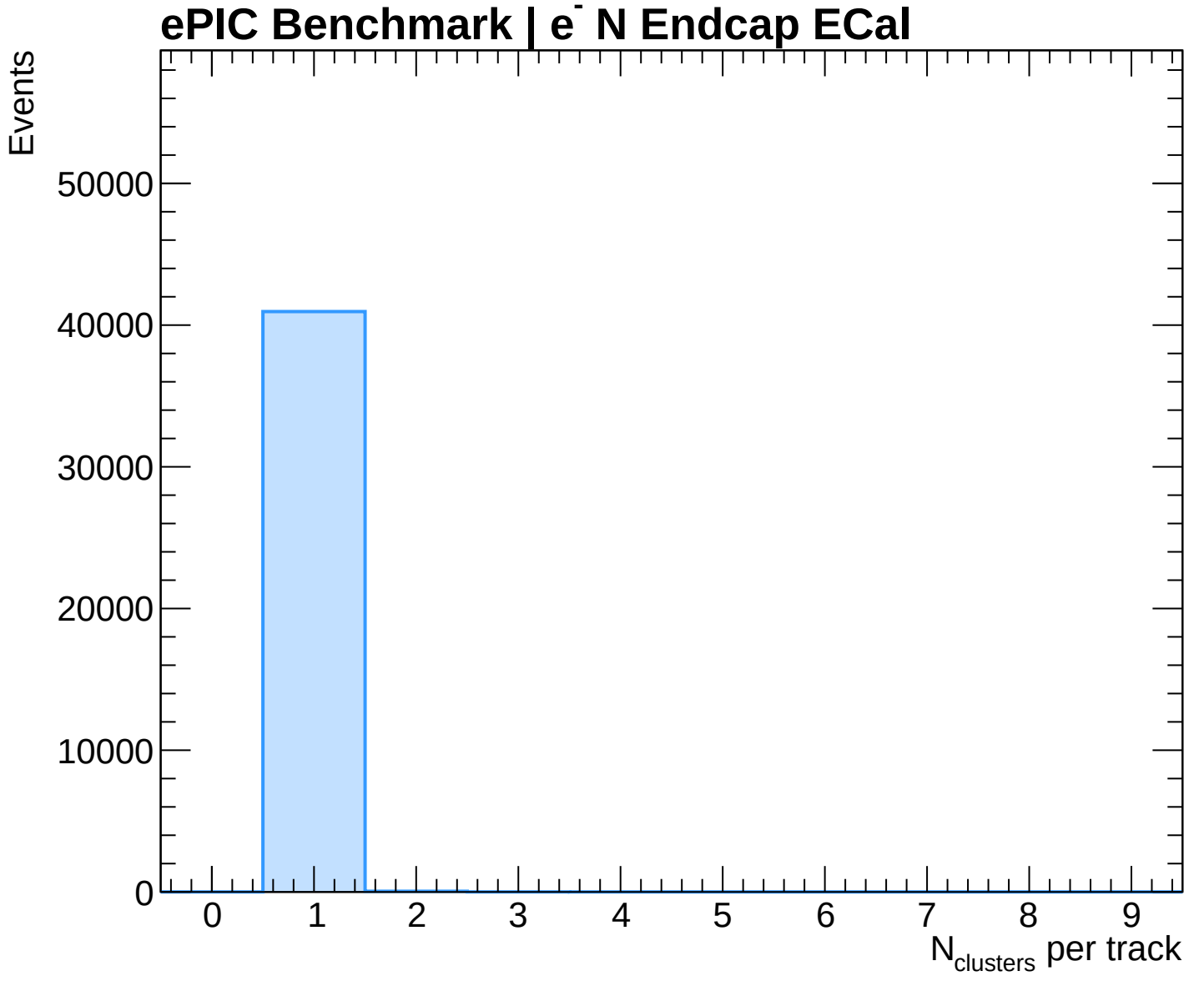

ePIC Benchmark | e- N Endcap ECal
Mean = 0.967
Std = 0.076
Entries = 41053
Events
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / E_Cl

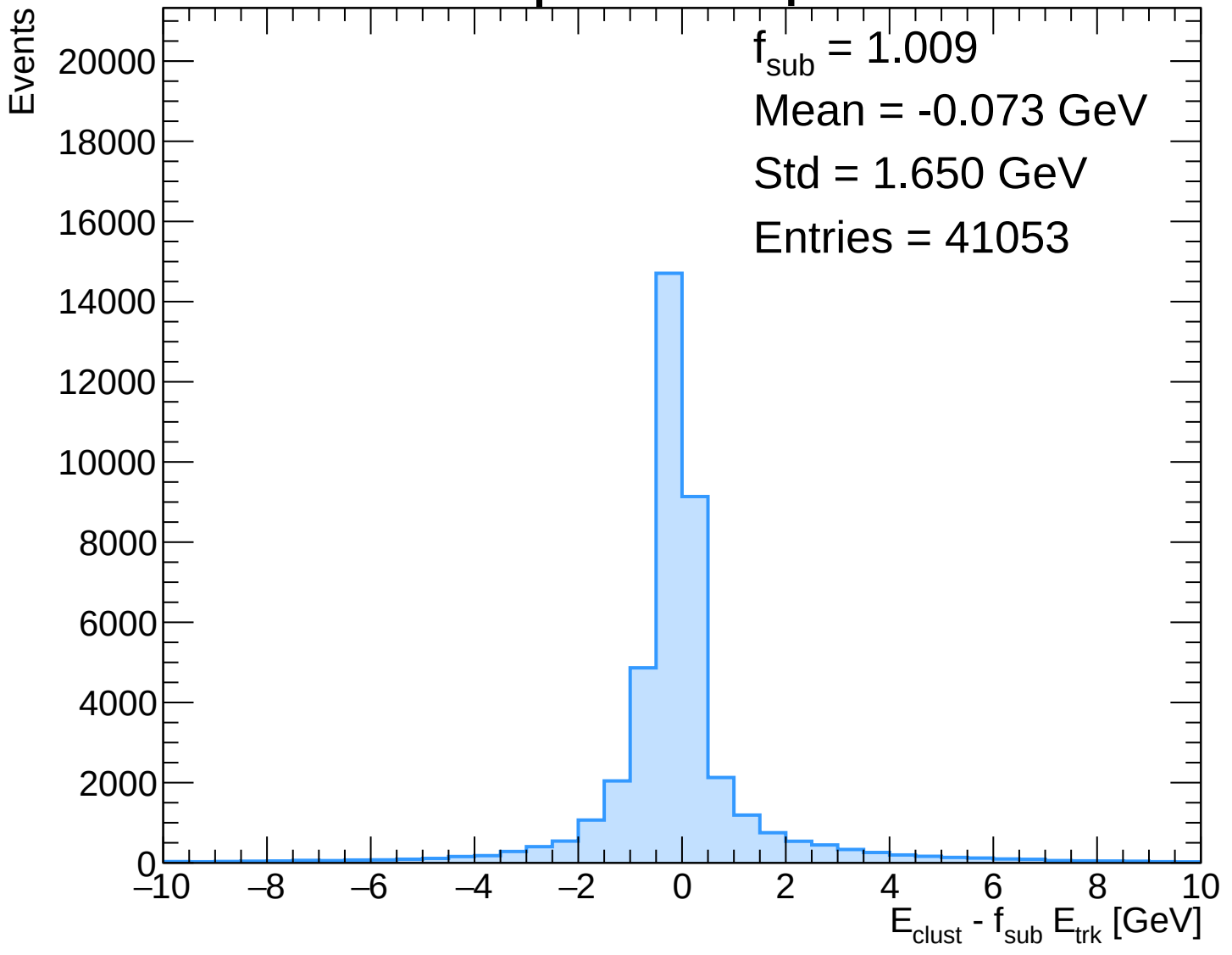

ePIC Benchmark | e- N Endcap ECal
Mean = 0.967
Std = 0.146
Entries = 41053
Events
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / p_Tr


ePIC Benchmark | e⁻ N Endcap ECal
Black: mean profile
f_sub = 1.009
E_Cl / E_Tr
Track p_T [GeV]
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0
1
2
3
4
5
6
7
0
100
200
300
400
500
600
700


ePIC Benchmark | e⁻ N Endcap ECal
Black: mean profile
f_sub = 1.009
E_Cl / E_Tr
Track η
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
–3.6
–3.4
–3.2
–3
–2.8
–2.6
–2.4
–2.2
–2
500
400
300
200
100
0


ePIC Benchmark | e- N Endcap ECal
Events
60000
50000
40000
30000
20000
10000
0
0
1
2
3
4
5
6
7
8
9
N_tracks per cluster


ePIC Benchmark | e- N Endcap ECal
Events
50000
40000
30000
20000
10000
0
0
1
2
3
4
5
6
7
8
9
N clusters per track


ePIC Benchmark | e- N Endcap ECal
Events
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
f_sub = 1.009
Mean = -0.073 GeV
Std = 1.650 GeV
Entries = 41053
–10
–8
–6
–4
–2
0
2
4
6
8
10
E_clust - f_sub E_trk [GeV]